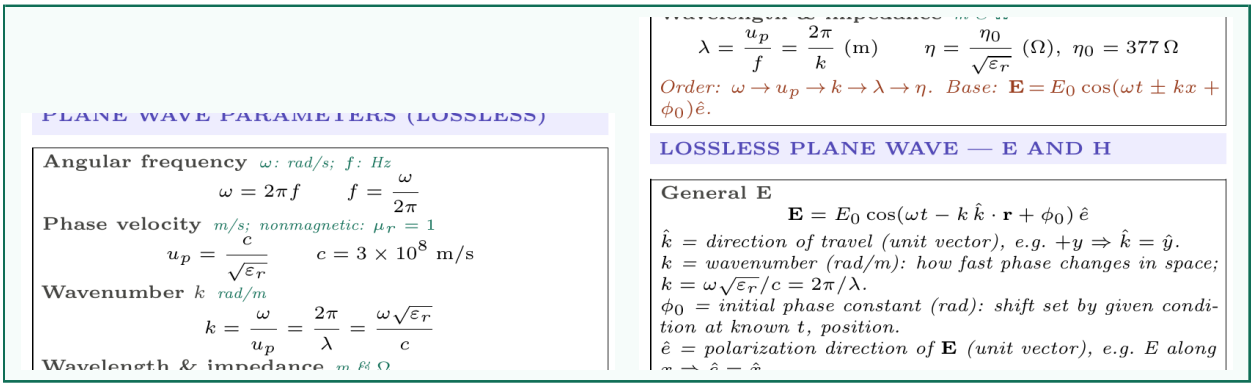
ECE 332 — Homework 2: Generated Answers from Cheat Sheet

Each solution shows the cheat sheet snippet used, then the answer.

Problem 1 — Plane Wave E and H with Phase Constant — 10 pts

Setup: 1 GHz wave traveling $+\hat{y}$ in lossless nonmagnetic medium with $\varepsilon_r = 9$. $\mathbf{E}$ along $\hat{x}$, peak 6 V/m, $|E| = 4$ V/m at $t = 0$, $y = 2$ cm.



PLANE WAVE PARAMETERS (LOSSLESS)

Angular frequency *ω: rad/s; f: Hz*

$$\omega = 2\pi f \qquad f = \frac{\omega}{2\pi}$$

Phase velocity *m/s; nonmagnetic: $\mu_r = 1$*

$$u_p = \frac{c}{\sqrt{\varepsilon_r}} \qquad c = 3 \times 10^8 \text{ m/s}$$

Wavenumber k *rad/m*

$$k = \frac{\omega}{u_p} = \frac{2\pi}{\lambda} = \frac{\omega\sqrt{\varepsilon_r}}{c}$$

Wavelength & impedance *m & Ω*

$$\lambda = \frac{u_p}{f} = \frac{2\pi}{k} \text{ (m)} \qquad \eta = \frac{\eta_0}{\sqrt{\varepsilon_r}} \text{ (Ω)}, \ \eta_0 = 377\,\Omega$$

Order: $\omega \to u_p \to k \to \lambda \to \eta$. Base: $\mathbf{E} = E_0 \cos(\omega t \pm kx + \phi_0)\hat{e}$.

LOSSLESS PLANE WAVE — E AND H

General E

$$\mathbf{E} = E_0 \cos(\omega t - k\,\hat{k}\cdot\mathbf{r} + \phi_0)\,\hat{e}$$

$\hat{k}$ = direction of travel (unit vector), e.g. $+y \Rightarrow \hat{k} = \hat{y}$.
k = wavenumber (rad/m): how fast phase changes in space; $k = \omega\sqrt{\varepsilon_r}/c = 2\pi/\lambda$.
ϕ_0 = initial phase constant (rad): shift set by given condition at known t, position.
$\hat{e}$ = polarization direction of $\mathbf{E}$ (unit vector), e.g. E along $x \to \hat{e} = \hat{x}$

Parameters from the cheat sheet order $\omega \to u_p \to k \to \lambda \to \eta$:

$$\omega = 2\pi f = 2\pi \times 10^9 \text{ rad/s}, \qquad k = \frac{\omega\sqrt{\varepsilon_r}}{c} = \frac{2\pi\times10^9\sqrt{9}}{3\times10^8} = 20\pi \text{ rad/m}$$

General E form (wave in $+\hat{y}$ so argument is $\omega t - ky$):

$$\mathbf{E}(y,t) = 6\cos(\omega t - ky + \phi_0)\,\hat{x} \text{ V/m}$$

Find ϕ_0 from the given condition $|E| = 4$ at $t = 0$, $y = 0.02$ m:

$$4 = 6\cos(-0.4\pi + \phi_0) \ \Rightarrow\ -0.4\pi + \phi_0 = \cos^{-1}(2/3) = 0.84 \text{ rad}$$

$$\boxed{\phi_0 = 2.1 \text{ rad}}$$

$$\boxed{\mathbf{E}(y,t) = 6\cos\big((2\pi\times10^9)t - (20\pi)y + 2.1\big)\,\hat{x} \text{ V/m}}$$

Magnetic field via $\mathbf{H} = \frac{1}{\eta}(\hat{k}\times\mathbf{E})$:

$$\eta = \frac{\eta_0}{\sqrt{\varepsilon_r}} = \frac{377}{3} \approx 125.7\ \Omega, \qquad \hat{y}\times\hat{x} = -\hat{z}$$

$$\boxed{\mathbf{H}(y,t) = -\frac{6}{125.7}\cos\big((2\pi\times10^9)t - (20\pi)y + 2.1\big)\,\hat{z} \text{ A/m}}$$

Problem 2 — f, u_p, λ, k, η and H — 12 pts

Setup: $\mathbf{E} = 20\cos((6\pi\times10^9)t - kz)\,\hat{y}$ V/m, lossless nonmagnetic, $\varepsilon_r = 2.56$.

PLANE WAVE PARAMETERS (LOSSLESS)

Angular frequency *ω: rad/s; f: Hz*

$$\omega = 2\pi f \qquad f = \frac{\omega}{2\pi}$$

Phase velocity *m/s; nonmagnetic: $\mu_r = 1$*

$$u_p = \frac{c}{\sqrt{\varepsilon_r}} \qquad c = 3\times10^8 \text{ m/s}$$

Wavenumber k *rad/m*

$$k = \frac{\omega}{u_p} = \frac{2\pi}{\lambda} = \frac{\omega\sqrt{\varepsilon_r}}{c}$$

Wavelength & impedance *m & Ω*

DIRECTION OF PROPAGATION

Arg. contains	$\hat{k}$	Arg. contains	$\hat{k}$
$\omega t - kz$	$+\hat{z}$	$\omega t + kz$	$-\hat{z}$
$\omega t - ky$	$+\hat{y}$	$\omega t + ky$	$-\hat{y}$
$\omega t - kx$	$+\hat{x}$	$\omega t + kx$	$-\hat{x}$

Minus before k-axis ⇒ + direction. Plus ⇒ − direction.

$\hat{k}\times\hat{E}$ — **FIND $\hat{H}$ DIRECTION**

$\hat{k}$	$\hat{E}$	$\hat{H}$	$\hat{k}$	$\hat{E}$	$\hat{H}$
$+\hat{z}$	$\hat{x}$	$+\hat{y}$	$+\hat{z}$	$\hat{y}$	$-\hat{x}$
$-\hat{z}$	$\hat{x}$	$-\hat{y}$	$-\hat{z}$	$\hat{y}$	$+\hat{x}$
$+\hat{y}$	$\hat{x}$	$-\hat{z}$	$+\hat{y}$	$\hat{z}$	$+\hat{x}$
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

(a) Apply the parameter chain in order:

$$\begin{aligned}
f &= \omega/(2\pi) = 3\times10^9 \text{ Hz}\\
u_p &= c/\sqrt{\varepsilon_r} = 3\times10^8/\sqrt{2.56} = 1.875\times10^8 \text{ m/s}\\
\lambda &= u_p/f = 6.25 \text{ cm}\\
k &= 2\pi/\lambda = 32\pi \approx 100.5 \text{ rad/m}\\
\eta &= \eta_0/\sqrt{\varepsilon_r} = 377/\sqrt{2.56} = 235.6\ \Omega
\end{aligned}$$

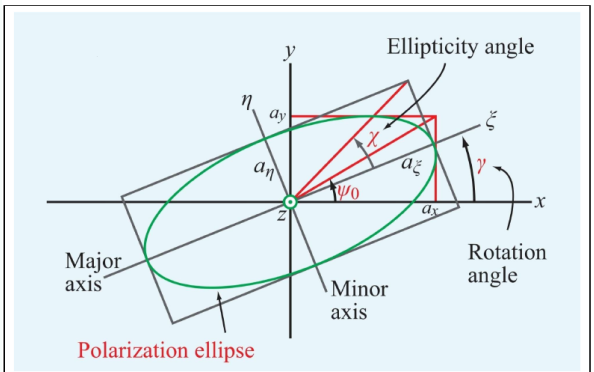
(b) H direction from the $\hat{k}\times\hat{E}$ table:
Wave in $+\hat{z}$, $\mathbf{E}$ along $\hat{y}$, so $\hat{z}\times\hat{y} = -\hat{x}$.

$$\boxed{\mathbf{H}(z,t) = -\tfrac{20}{235.6}\cos\big((6\pi\times10^9)t - (32\pi)z\big)\,\hat{x} = -0.0849\cos(\cdot)\,\hat{x}\ \text{A/m}}$$

Problem 3 — Polarization Analysis (4 cases) 16 pts

Setup: $\mathbf{E}(z,t) = a_x \cos(\omega t - kz)\hat{x} + a_y \cos(\omega t - kz + \delta)\hat{y}$.

POLARIZATION — GENERAL WAVE



$\mathbf{E} = a_x \cos(\omega t - kz)\hat{x} + a_y \cos(\omega t - kz + \delta)\hat{y}$

a_x, a_y: bounding box half-widths (V/m); δ: phase diff (rad)

$\delta = 0$: $\mathbf{E} = (a_x\hat{x} + a_y\hat{y})\cos(\omega t - kz)$ *(linear)*

$\delta = \pi$: $\mathbf{E} = (a_x\hat{x} - a_y\hat{y})\cos(\omega t - kz)$ *(linear)*

Auxiliary angle $\psi_0 = \arctan(a_y/a_x) \in (0, \pi/2)$

χ describes the shape of the ellipse and rotation direction.

χ (rad)	Polarization Type	Handedness
0	Linear	None
$(0, \pi/4)$	RHEP	RH (CCW)
$\pi/4$	RHC	RH (CCW)
$-\pi/4$	LHC	LH (CW)
$(-\pi/4, 0)$	LHEP	LH (CW)

QUADRANT RULE FOR $\tan(2\gamma)$

$N = \sin(2\psi_0)\cos\delta, \quad D = \cos(2\psi_0)$:

Rotation angle γ **(major axis from** $\hat{x}$**)** $\gamma \in [0, \pi)$

$$\tan(2\gamma) = \tan(2\psi_0)\cos\delta$$

If sign(γ) $\neq$ sign(δ): $\gamma \leftarrow \gamma \pm \frac{\pi}{2}$.

Ellipticity angle χ $\chi \in [-\pi/4, \pi/4]$

$$\sin(2\chi) = \sin(2\psi_0)\sin\delta$$

Axial ratio $R = 1/\tan|\chi|$ *major axis = $R\times$ minor; $R \geq 1$*

χ SHAPE & HANDEDNESS

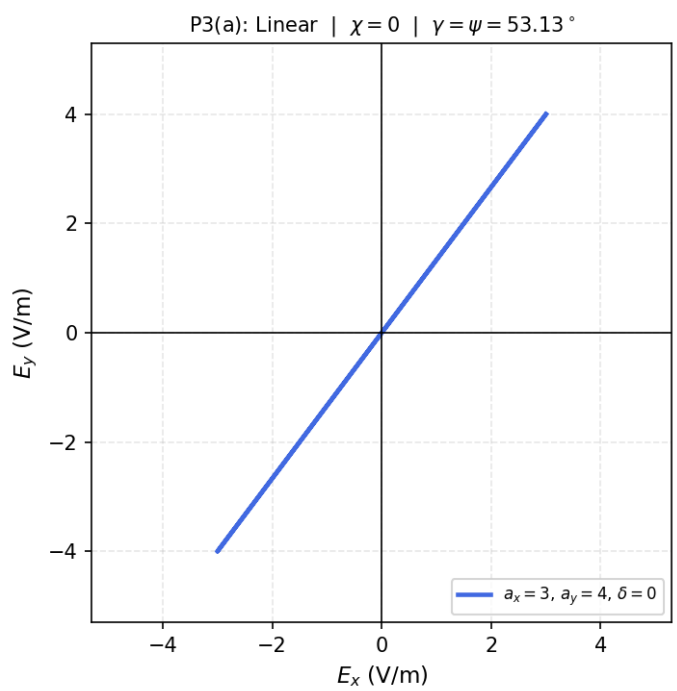
A	$\mathrm{Re}\{Ae^{j(\omega t-\beta z)}\}$	$A\cos(\omega t-\beta z)$
$-jA$	$\mathrm{Re}\{Ae^{j(\omega t-\beta z-\pi/2)}\}$	$A\sin(\omega t-\beta z)$
$+jA$	$\mathrm{Re}\{Ae^{j(\omega t-\beta z+\pi/2)}\}$	$-A\sin(\omega t-\beta z)$
$-A$	$\mathrm{Re}\{Ae^{j(\omega t-\beta z\pm\pi)}\}$	$-A\cos(\omega t-\beta z)$
$Ae^{j\phi}$	$\mathrm{Re}\{Ae^{j(\omega t-\beta z+\phi)}\}$	$A\cos(\omega t-\beta z+\phi)$
$A+jB$	$\mathrm{Re}\{\lvert Z\rvert e^{j(\omega t-\beta z+\theta)}\}$	$\lvert Z\rvert\cos(\omega t-\beta z+\theta)$

$|Z| = \sqrt{A^2+B^2}$, $\theta = \arctan(B/A)$ (check quadrant)

CIRCULAR POLARIZATION — TIME DOMAIN

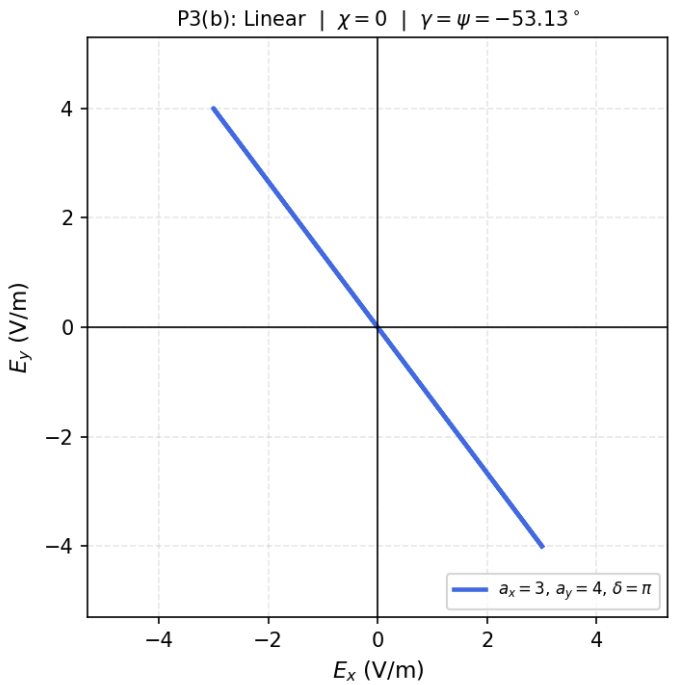
(a) $a_x = 3$, $a_y = 4$, $\delta = 0$**:** $\delta = 0 \Rightarrow$ linear. $R = \infty$, $\chi = 0$. Rotation angle = inclination:

$$\psi = \tan^{-1}(4/3) = 0.9273 \text{ rad} = 53.13^\circ \Rightarrow \boxed{\gamma = 53.13^\circ,\ \chi = 0}$$



(b) $a_x = 3$**,** $a_y = 4$**,** $\delta = \pi$**:** $\delta = \pi \Rightarrow$ linear again ($\cos(\theta+\pi) = -\cos\theta$, so $\mathbf{E} = (3\hat{x}-4\hat{y})\cos(\omega t - kz)$). $\chi = 0$.

$$\psi = \tan^{-1}(-4/3) = -0.9273 \text{ rad} = -53.13^\circ \Rightarrow \boxed{\gamma = -53.13^\circ,\ \chi = 0}$$



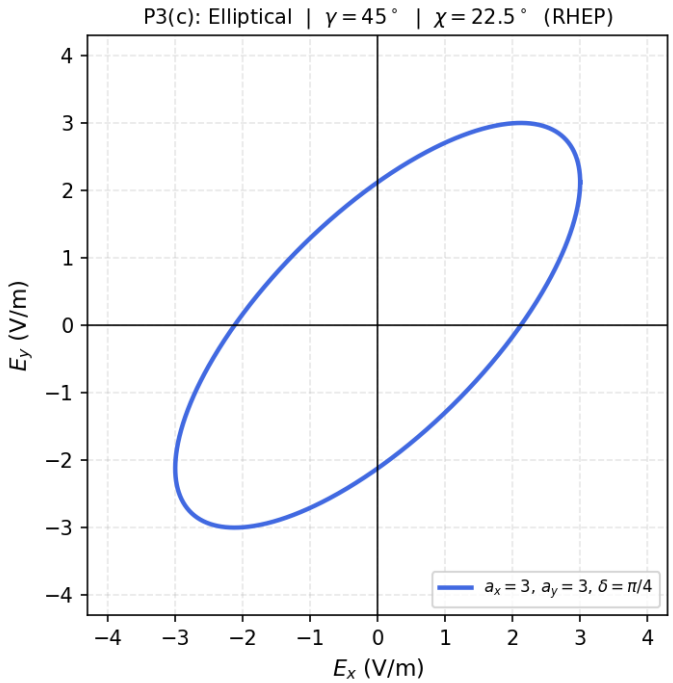
(c) $a_x = 3$**,** $a_y = 3$**,** $\delta = \pi/4$**:** elliptical. Auxiliary $\psi_0 = \tan^{-1}(1) = \pi/4$.

$$\tan 2\gamma = \tan(\pi/2)\cos(\pi/4) = \infty\sqrt{2}/2 \Rightarrow 2\gamma = \pi/2$$

$$\sin 2\chi = \sin(\pi/2)\sin(\pi/4) = \sqrt{2}/2 \Rightarrow 2\chi = \pi/4$$

$$\boxed{\gamma = \pi/4 = 45^\circ, \quad \chi = \pi/8 = 22.5^\circ}$$

$\chi > 0 \Rightarrow$ right-hand elliptical (RHEP).



(d) $a_x = 3$**,** $a_y = 4$**,** $\delta = -3\pi/4$**:** elliptical. $\psi_0 = \tan^{-1}(4/3) \approx 0.9273$, $2\psi_0 \approx 1.8546$ (106.26°).

$$\cos\delta = -\sqrt{2}/2, \quad \sin\delta = -\sqrt{2}/2$$

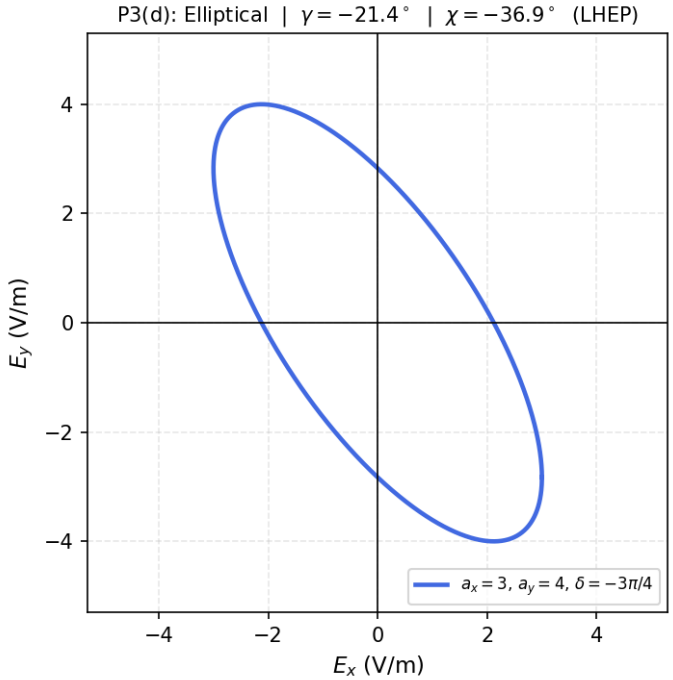
$$\tan 2\gamma = \tan(1.8546) \cdot (-\sqrt{2}/2) \approx 2.4244$$

$N = \sin(2\psi_0)\cos\delta < 0$, $D = \cos(2\psi_0) < 0$ (quadrant rule: both negative)

$$2\gamma \approx -1.1796 \text{ rad} = -73.73^\circ \Rightarrow \boxed{\gamma \approx -21.4^\circ}$$

$$\sin 2\chi = \sin(1.8546) \cdot (-\sqrt{2}/2) = -0.7462 \Rightarrow 2\chi \approx -0.7462 \text{ rad} \Rightarrow \boxed{\chi \approx -36.9^\circ}$$

$\chi < 0 \Rightarrow$ left-hand elliptical (LHEP).


P3(d): Elliptical | $\gamma = -21.4^\circ$ | $\chi = -36.9^\circ$ (LHEP)
E_y (V/m)
E_x (V/m)
4
2
0
−2
−4
−4
−2
0
2
4
$a_x = 3,\ a_y = 4,\ \delta = -3\pi/4$

Problem 4 — Linear = RHC + LHC Decomposition 8 pts

Setup: Show $\tilde{\mathbf{E}} = a_x e^{-jkz}\hat{x}$ can be written as $\tilde{\mathbf{E}}_R + \tilde{\mathbf{E}}_L$.

A	$\mathrm{Re}\{Ae^{j(\omega t-\beta z)}\}$	$A\cos(\omega t-\beta z)$
$-jA$	$\mathrm{Re}\{Ae^{j(\omega t-\beta z-\pi/2)}\}$	$A\sin(\omega t-\beta z)$
$+jA$	$\mathrm{Re}\{Ae^{j(\omega t-\beta z+\pi/2)}\}$	$-A\sin(\omega t-\beta z)$
$-A$	$\mathrm{Re}\{Ae^{j(\omega t-\beta z\pm\pi)}\}$	$-A\cos(\omega t-\beta z)$
$Ae^{j\phi}$	$\mathrm{Re}\{Ae^{j(\omega t-\beta z+\phi)}\}$	$A\cos(\omega t-\beta z+\phi)$
$A+jB$	$\mathrm{Re}\{\lvert Z\rvert e^{j(\omega t-\beta z+\theta)}\}$	$\lvert Z\rvert\cos(\omega t-\beta z+\theta)$

$|Z| = \sqrt{A^2+B^2}$, $\theta = \arctan(B/A)$ (check quadrant)

CIRCULAR POLARIZATION — TIME DOMAIN

From the cheat sheet definitions:

$$\tilde{\mathbf{E}}_R = a_R(\hat{x} - j\hat{y})e^{-jkz}, \qquad \tilde{\mathbf{E}}_L = a_L(\hat{x} + j\hat{y})e^{-jkz}$$

Try $a_R = a_L = a_x/2$**:**

$$\begin{aligned}\tilde{\mathbf{E}}_R + \tilde{\mathbf{E}}_L &= \big[(a_R + a_L)\hat{x} + j(-a_R + a_L)\hat{y}\big]e^{-jkz} \\ &= \big[a_x\,\hat{x} + j\cdot 0\cdot\hat{y}\big]e^{-jkz} = a_x e^{-jkz}\hat{x} = \tilde{\mathbf{E}}\ \checkmark\end{aligned}$$

$$\boxed{a_R = a_L = a_x/2}$$

Problem 5 — Dry Soil at 4 Frequencies (Loss Tangent Classification) 24 pts

Setup: $\varepsilon_r = 2.5$, $\mu_r = 1$, $\sigma = 10^{-4}$ S/m.

LOSSY MEDIA — CLASSIFICATION

Loss tangent

$$\frac{\varepsilon''}{\varepsilon'} = \frac{\sigma}{\omega\varepsilon_r\varepsilon_0}$$

$\varepsilon''/\varepsilon'$	Type	Use
> 100	Good conductor	GC approx
0.01 to 100	Quasi-conductor	Exact eqs
< 0.01	Low-loss dielectric	LD approx

$\varepsilon' = \varepsilon_r\varepsilon_0$; $\varepsilon'' = \sigma/\omega$; *nonmag.:* $\mu = \mu_0$; *free space:* $\varepsilon_r = \mu_r = 1$.

ANY MEDIUM — EXACT ($X = \sqrt{1+(\varepsilon''/\varepsilon')^2}$)

$$\alpha = \omega\sqrt{\frac{\mu\varepsilon'}{2}}\sqrt{X-1}\ \text{Np/m}, \quad \beta = \omega\sqrt{\frac{\mu\varepsilon'}{2}}\sqrt{X+1}\ \text{rad/m}$$

$$\lambda = \frac{2\pi}{\beta} = \frac{u_p}{f}\ (\text{m}), \quad u_p = \frac{\omega}{\beta}\ (\text{m/s})$$

$$\eta_c = \frac{\eta}{X^{1/2}}e^{j\theta_\eta}\ (\Omega), \quad \theta_\eta = \frac{1}{2}\arctan\frac{\sigma}{\omega\varepsilon'}\ (\text{rad})$$

SKIN DEPTH δ_s (m) — ANY LOSSY MEDIUM

General: $\delta_s = 1/\alpha$ $|E(z)| = E_0 e^{-z/\delta_s}$; *lossless:* $\alpha = 0 \Rightarrow$ *no decay*
Low-loss ($\varepsilon''/\varepsilon' \ll 1$) $\delta_s \approx 2/(\sigma\eta)$
Good conductor ($\varepsilon''/\varepsilon' \gg 1$) $\delta_s \approx 1/\sqrt{\pi f\mu\sigma}$

LOW-LOSS MEDIUM ($\varepsilon''/\varepsilon' \ll 1$)

$$\alpha \approx \frac{\sigma}{2}\sqrt{\frac{\mu}{\varepsilon'}} = \frac{\sigma\eta}{2}\ (\text{Np/m})$$

$$\beta \approx \omega\sqrt{\mu\varepsilon'} = \frac{\omega\sqrt{\varepsilon_r}}{c}\ (\text{rad/m}), \quad u_p \approx \frac{c}{\sqrt{\varepsilon_r}}\ (\text{m/s})$$

$$\eta_c \approx \sqrt{\frac{\mu}{\varepsilon'}} = \eta\ (\Omega,\ \angle \approx 0)$$

Compute the loss tangent ratio:

$$\frac{\varepsilon''}{\varepsilon'} = \frac{\sigma}{\omega\varepsilon_r\varepsilon_0} = \frac{\sigma/(2\pi\varepsilon_r\varepsilon_0)}{f} = \frac{7.193\times 10^5\ \text{Hz}}{f}$$

f	$\varepsilon''/\varepsilon'$	Class	Use
60 Hz	1.20×10^4	Good conductor	$\alpha \approx \beta \approx \sqrt{\pi f \mu \sigma}$
1 kHz	719.3	Good conductor	GC approx
1 MHz	0.719	Quasi-conductor	Exact eqs
1 GHz	7.19×10^{-4}	Low-loss	LD approx

(a) 60 Hz (good conductor):

$$\alpha \approx \beta = \sqrt{\pi(60)(4\pi \times 10^{-7})(10^{-4})} = 4.87 \times 10^{-6} \text{ Np/m, rad/m}$$

$$\lambda = 2\pi/\beta = 1.29 \times 10^6 \text{ m}, \quad u_p = \sqrt{4\pi f/\mu\sigma} = 7.74 \times 10^7 \text{ m/s}$$

$$\eta_c \approx (1+j)\alpha/\sigma = (1+j)(0.0487)\ \Omega$$

(b) 1 kHz (good conductor): same formulas, scale by $\sqrt{1000/60} \approx 4.08\times$:

$$\alpha \approx \beta = 1.99 \times 10^{-5} \text{ Np/m}, \quad \lambda \approx 3.16 \times 10^5 \text{ m}$$

(c) 1 MHz (quasi-conductor): use exact $X = \sqrt{1 + (\varepsilon''/\varepsilon')^2} = \sqrt{1 + 0.719^2} = 1.231$:

$$\alpha = \omega\sqrt{\tfrac{\mu\varepsilon'}{2}}\sqrt{X-1} \text{ Np/m}, \quad \beta = \omega\sqrt{\tfrac{\mu\varepsilon'}{2}}\sqrt{X+1} \text{ rad/m}$$

$$\eta_c = \tfrac{\eta}{\sqrt{X}} e^{j\theta_\eta}, \quad \theta_\eta = \tfrac{1}{2}\arctan(0.719) = 17.7^\circ$$

(d) 1 GHz (low-loss dielectric):

$$\alpha \approx \tfrac{\sigma}{2}\sqrt{\mu/\varepsilon'} = \tfrac{10^{-4}}{2} \cdot \tfrac{377}{\sqrt{2.5}} = 0.0119 \text{ Np/m}$$

$$\beta \approx \omega\sqrt{\mu\varepsilon'} = \tfrac{2\pi \times 10^9 \sqrt{2.5}}{3 \times 10^8} = 33.1 \text{ rad/m}$$

$$\lambda \approx 2\pi/\beta = 0.190 \text{ m}, \ u_p \approx c/\sqrt{\varepsilon_r} = 1.90 \times 10^8 \text{ m/s}, \ \eta_c \approx 238.4\ \Omega$$

Problem 6 — Phasor → Time Domain (Lossy) 10 pts

Setup: 300 MHz, nonmagnetic lossy, $\tilde{\mathbf{H}} = (\hat{x} - j4\hat{z})e^{-2y}e^{-j9y}$ A/m.

N	D	2γ (deg)	2γ (rad)
$+$	$+$	$\arctan(N/D)$	$\arctan(N/D)$
$+$	$-$	$180^\circ - \arctan\lvert N/D\rvert$	$\pi - \arctan\lvert N/D\rvert$
$-$	$-$	$180^\circ + \arctan\lvert N/D\rvert$	$\pi + \arctan\lvert N/D\rvert$
$-$	$+$	$360^\circ - \arctan\lvert N/D\rvert$	$2\pi - \arctan\lvert N/D\rvert$

$\gamma = (2\gamma)/2$. $D = 0 \Rightarrow 2\gamma = \frac{\pi}{2} \Rightarrow \gamma = \frac{\pi}{4}$.
Convert: $\theta_{rad} = \theta_\circ \times \frac{\pi}{180}$; $\theta_\circ = \theta_{rad} \times \frac{180}{\pi}$.

PHASOR ↔ TIME DOMAIN

Phasor coeff	$\mathrm{Re}\{(\cdot)\, e^{j\omega t}e^{-j\beta z}\}$	Result

DIRECTION OF PROPAGATION

Arg. contains	$\hat{k}$	Arg. contains	$\hat{k}$
$\omega t - kz$	$+\hat{z}$	$\omega t + kz$	$-\hat{z}$
$\omega t - ky$	$+\hat{y}$	$\omega t + ky$	$-\hat{y}$
$\omega t - kx$	$+\hat{x}$	$\omega t + kx$	$-\hat{x}$

Minus before k-axis ⇒ + direction. Plus ⇒ − direction.

$\hat{k} \times \hat{E}$ — FIND $\hat{H}$ DIRECTION

$\hat{k}$	$\hat{E}$	$\hat{H}$	$\hat{k}$	$\hat{E}$	$\hat{H}$
$+\hat{z}$	$\hat{x}$	$+\hat{y}$	$+\hat{z}$	$\hat{y}$	$-\hat{x}$
$-\hat{z}$	$\hat{x}$	$-\hat{y}$	$-\hat{z}$	$\hat{y}$	$+\hat{x}$
$+\hat{y}$	$\hat{x}$	$-\hat{z}$	$+\hat{y}$	$\hat{z}$	$+\hat{x}$
$-\hat{y}$	$\hat{x}$	$+\hat{z}$	$-\hat{y}$	$\hat{z}$	$-\hat{x}$

Read off: $\omega = 6\pi \times 10^8$ rad/s, $\alpha = 2$ Np/m, $\beta = 9$ rad/m, $\hat{k} = \hat{y}$.

Time-domain H using the phasor table ($-j \to \sin$):

$$\mathbf{H}(y,t) = \big[\cos(\omega t - 9y)\,\hat{x} \;-\; 4\cos(\omega t - 9y + \tfrac{\pi}{2})\,\hat{z}\big]e^{-2y} \text{ A/m}$$

Find $\tilde{\mathbf{E}}$ using $\tilde{\mathbf{E}} = -\eta_c(\hat{k} \times \tilde{\mathbf{H}})$, but for lossy we need η_c first. Compute loss tangent:

$$\varepsilon''/\varepsilon' = (\alpha\beta \cdot 2)/(\omega^2\mu\varepsilon') \quad \text{(implicit from } \alpha, \beta)$$

Or compute $\eta_c = j\omega\mu_0/\gamma$ with $\gamma = \alpha + j\beta = 2 + j9$:

$$\eta_c = \frac{j(6\pi\times10^8)(4\pi\times10^{-7})}{2+j9} = \frac{j(753.98)}{2+j9} = \frac{j753.98\,(2-j9)}{4+81} = \frac{6785.8+j1507.9}{85} \approx 79.8 + j17.7\ \Omega$$

Then $\hat{y} \times \hat{x} = -\hat{z}$, $\hat{y} \times \hat{z} = \hat{x}$:

$$\tilde{\mathbf{E}} = -\eta_c\big(\hat{y} \times (\hat{x} - j4\hat{z})\big)e^{-2y}e^{-j9y} = -\eta_c(-\hat{z} - j4\hat{x})e^{-2y}e^{-j9y}$$

$$\tilde{\mathbf{E}} = \eta_c(j4\hat{x} + \hat{z})e^{-2y}e^{-j9y} \text{ V/m}$$

Time domain:

$$\mathbf{E}(y,t) = |\eta_c|\big[4\cos(\omega t - 9y + \pi/2 + \theta_\eta)\,\hat{x} + \cos(\omega t - 9y + \theta_\eta)\,\hat{z}\big]e^{-2y}$$

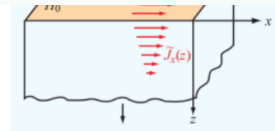
where $|\eta_c| \approx 81.8$, $\theta_\eta = \arctan(17.7/79.8) \approx 12.5^\circ$.

Problem 7 — Direction of Travel and Poynting 10 pts

Setup: $\mathbf{E} = 3\cos(\pi\times10^7 t + kx)\hat{y} - 2\cos(\pi\times10^7 t + kx)\hat{z}$ V/m, nonmagnetic $\varepsilon_r = 9$.

Lossless shortcuts (wave in $+z$, $\mathbf{E}$ along $\hat{e}$):

$\mathbf{E}(z,t)$	$E_0 \cos(\omega t - kz + \phi)\,\hat{e}$
$\mathbf{H}(z,t)$	$\frac{E_0}{\eta} \cos(\omega t - kz + \phi)\,(\hat{k} \times \hat{e})$
$\mathbf{H}(t)$ direct	$\frac{1}{\eta}(\hat{k} \times \mathbf{E}(t))$
$\mathbf{E}(t)$ direct	$-\eta(\hat{k} \times \mathbf{H}(t))$



(a) Exponentially decaying $\tilde{J}_x(z)$

$J_0 = \sigma E_0$ — current density at surface (z=0)
$\tilde{\mathbf{J}}(z) = J_0 e^{-(1+j)z/\delta_s}\,\hat{x}$ — decays with depth
$R_s = \sqrt{\pi f \mu/\sigma} = 1/(\sigma\delta_s)$ (Ω)
$Z_s = (1+j)R_s$ — surface impedance
$R_{\mathrm{AC}} = R_s\,\ell/w$ — AC resistance of slab

AVERAGE POWER DENSITY (POYNTING)

Direction from the table: argument is $\omega t + kx$ (plus sign $\Rightarrow$ wave travels in $-\hat{x}$).

$$\boxed{\hat{k} = -\hat{x}}$$

No $e^{-\alpha x}$ attenuation factor, so the medium is effectively lossless here.

Intrinsic impedance:

$$\eta = \eta_0/\sqrt{\varepsilon_r} = 377/3 \approx 125.7\ \Omega = 40\pi\ \Omega$$

Poynting (lossless shortcut): $|E_0|^2 = 3^2 + 2^2 = 13$.

$$\mathbf{S}_{\mathrm{av}} = \frac{|E_0|^2}{2\eta}\hat{k} = \frac{13}{80\pi}(-\hat{x})\ \mathrm{W/m^2} \approx -51.7\ \mathrm{mW/m^2}\,\hat{x}$$

$$\boxed{\mathbf{S}_{\mathrm{av}} \approx -51.7\ \mathrm{mW/m^2}\ \hat{x}}$$